(2.) Evaluating RAG solutions →

(V1) → RAG Evaluation Introduction →

→ Evaluating RAG system solution with Ragas.

→ Ragas → Ragas helps to evaluate the system in a structured way.

→ Issues while using RAG →

(1) Retrieval error →
when we try to retrieve information → and the text ~~genera~~ retrieved is not relevant

(2) Generation Error →
The response generated, but not mentioned anywhere in the document is called generation error.

→ Topics included →

(1) Why rag pipeline can fail and why? →
evaluating them requires targeted metrics for retreival and generation error.

(2) How to setup evaluation environment

(3) Full evaluation with Ragas → and to go through retrival and generational metrics in details, then interpreting the outputs and analyzing ethical risks like harmfulness and maliciousness flagged using score.

→ Evaluation Metrics (Ruga) →

(1) Retrieval metrics : Context Precision @ k
→ here k represents the no of allowed docs to be retrieved and defines how many context generator sees.

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→ measures → how many of the retrieved documents are relevant to the question?

② Retrieval metrics: context Recall @ K.

→ It defines → "Were all the relevant documents retrieved?"

→ It calculates the number of relevant document retrieved divided by the estimated top number of relevant document and returns score → 0 and 1.

③ Generation metrics → faithfullness →

→ It defines whether "the generated answer is supported by the retrieved documents?"

→ If the answer includes a claim not founded in the retrieved content, faithfullness drop

→ It is scored between zero and one

→ It is low then it means the answer is ~~tipsy~~ unsupported.

→ It detects hallucination.

④ Generation metrics → Response Relevance →

→ It defines "Is the response relevant to the user input"?"

→ It ~~addresses~~ measures whether the generated answer actually addres the input query.

→ here we can use ragas →

→ so Ragas uses llm to generate follow up question based on the ~~ans~~ answer

→ Then compares the similarity to the original query

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⑤ Generation metrics → Answer correctness →

→ It define = "Is the generated answer accurate when compared to the ground truth?"

→ It checks the factual accuracy of the answer against the reference answer independently of the retrieved context.

⑥ Generation metrics → Aspect Critique →

→ It defines → " Is the response is alligned with a predefined aspects, such as coherence ?"

→ This also uses Ragas →

→ the aspect critique is an LLM that takes the answer and the definition of the aspect in this case.

→ It generates binary output that is yes or no

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